

Discovering governing equations from data by sparse identification of nonlinear dynamical systems

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Extracting governing equations from data is a central challenge in many diverse areas of science and engineering. Data are abundant whereas models often remain elusive, as in climate science, neuroscience, ecology, finance, and epidemiology, to name only a few examples. In this work, we combine sparsity-promoting techniques and machine learning with nonlinear dynamical systems to discover governing equations from noisy measurement data. The only assumption about the structure of the model is that there are only a few important terms that govern the dynamics, so that the equations are sparse in the space of possible functions; this assumption holds for many physical systems in an appropriate basis. In particular, we use sparse regression to determine the fewest terms in the dynamic governing equations required to accurately represent the data. This results in parsimonious models that balance accuracy with model complexity to avoid overfitting. We demonstrate the algorithm on a wide range of problems, from simple canonical systems, including linear and nonlinear oscillators and the chaotic Lorenz system, to the fluid vortex shedding behind an obstacle. The fluid example illustrates the ability of this method to discover the underlying dynamics of a system that took experts in the community nearly 30 years to resolve. We also show that this method generalizes to parameterized systems and systems that are time-varying or have external forcing.

dynamical systems | machine learning | sparse regression | system identification | optimization

Advances in machine learning (1) and data science (2) have promised a renaissance in the analysis and understanding of complex data, extracting patterns in vast multimodal data that are beyond the ability of humans to grasp. However, despite the rapid development of tools to understand static data based on statistical relationships, there has been slow progress in distilling physical models of dynamic processes from big data. This has limited the ability of data science models to extrapolate the dynamics beyond the attractor where they were sampled and constructed.

An analogy may be drawn with the discoveries of Kepler and Newton. Kepler, equipped with the most extensive and accurate planetary data of the era, developed a data-driven model for planetary motion, resulting in his famous elliptic orbits. However, this was an attractor-based view of the world, and it did not explain the fundamental dynamic relationships that give rise to planetary orbits, or provide a model for how these bodies react when perturbed. Newton, in contrast, discovered a dynamic relationship between momentum and energy that described the underlying processes responsible for these elliptic orbits. This dynamic model may be generalized to predict behavior in regimes where no data were collected. Newton's model has proven remarkably robust for engineering design, making it possible to land a spacecraft on the moon, which would not have been possible using Kepler's model alone.

A seminal breakthrough by Bongard and Lipson (3) and Schmidt and Lipson (4) has resulted in a new approach to determine the underlying structure of a nonlinear dynamical system from data. This method uses symbolic regression [i.e., genetic programming (5)] to find nonlinear differential equations, and it balances complexity of the model, measured in the number of terms, with model accuracy. The resulting model identification realizes a long-sought goal of the physics and engineering communities to discover

dynamical systems from data. However, symbolic regression is expensive, does not scale well to large systems of interest, and may be prone to overfitting unless care is taken to explicitly balance model complexity with predictive power. In ref. 4, the Pareto front is used to find parsimonious models. There are other techniques that address various aspects of the dynamical system discovery problem. These include methods to discover governing equations from time-series data (6), equation-free modeling (7), empirical dynamic modeling (8, 9), modeling emergent behavior (10), and automated inference of dynamics (11–13); ref. 12 provides an excellent review.

Sparse Identification of Nonlinear Dynamics (SINDy)

In this work, we envision the dynamical system discovery problem from the perspective of sparse regression (14–16) and compressed sensing (17–22). In particular, we leverage the fact that most physical systems have only a few relevant terms that define the dynamics, making the governing equations sparse in a high-dimensional nonlinear function space. The combination of sparsity methods in dynamical systems is quite recent (23–30). Here, we consider dynamical systems (31) of the form

$$\frac{d}{dt}\mathbf{x}(t) = \mathbf{f}(\mathbf{x}(t)). \quad [1]$$

The vector $\mathbf{x}(t) \in \mathbb{R}^n$ denotes the state of a system at time t , and the function $\mathbf{f}(\mathbf{x}(t))$ represents the dynamic constraints that define the equations of motion of the system, such as Newton's second law. Later, the dynamics will be generalized to include parameterization, time dependence, and forcing.

Significance

Understanding dynamic constraints and balances in nature has facilitated rapid development of knowledge and enabled technology, including aircraft, combustion engines, satellites, and electrical power. This work develops a novel framework to discover governing equations underlying a dynamical system simply from data measurements, leveraging advances in sparsity techniques and machine learning. The resulting models are parsimonious, balancing model complexity with descriptive ability while avoiding overfitting. There are many critical data-driven problems, such as understanding cognition from neural recordings, inferring climate patterns, determining stability of financial markets, predicting and suppressing the spread of disease, and controlling turbulence for greener transportation and energy. With abundant data and elusive laws, data-driven discovery of dynamics will continue to play an important role in these efforts.

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通过对非线性动力系统进行稀疏识别从数据中发现控制方程

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从数据中提取控制方程是许多不同科学和工程领域中的一个核心挑战。数据丰富, 而模型通常难以获得, 例如在气候科学、神经科学、生态学、金融和流行病学中, 仅举几个例子。在这项工作中, 我们将稀疏性促进技术和机器学习与非线性动力系统相结合, 从噪声测量数据中发现控制方程。关于模型结构的唯一假设是只有少数重要项控制动力学, 因此在可能的函数空间中方程是稀疏的; 在适当的基下, 这一假设适用于许多物理系统。特别地, 我们使用稀疏回归来确定动态控制方程中所需的最少项, 以准确表示数据。这产生了简约模型, 在准确性与模型复杂性之间取得平衡, 以避免过拟合。我们在广泛的场景中演示了该算法

从数据中研究动力系统。然而, 符号回归成本高, 不适合扩展到大型感兴趣系统, 并且除非特别注意明确平衡模型复杂性与预测能力, 否则可能容易过拟合。在参考文献4中, 使用帕累托前沿来寻找简约模型。还有其他技术处理动力系统发现问题的各个方面。这些包括从时间序列数据中发现控制方程的方法(6)、无方程建模(7)、经验动力学建模(8, 9)、建模涌现行为(10)以及动态的自动推断(11–13); 参考文献12提供了出色的综述。

非线性动力学的稀疏识别 (SINDy)

在这项工作中, 我们从稀疏回归 (14–16) 和压缩感知 (17–22) 的角度重新审视动力系统发现问题。特别地, 我们利用这样一个事实: 大多数物理系统只有少数相关项能够定义其动态, 使得控制方程在高维非线性函数空间中是稀疏的。在动力系统中结合稀疏方法是相当新的 (23–30)。在这里, 我们考虑形式为的动力系统 (31)

$$\frac{d}{dt}\mathbf{x}(t) = \mathbf{f}(\mathbf{x}(t)). \quad [1]$$

向量 $\mathbf{x}(t) \in \mathbb{R}^n$ 表示系统在时间 t 的状态, 函数 $\mathbf{f}(\mathbf{x})$ 表示定义系统运动方程的动态约束, 例如牛顿第二定律。随后, 动力学将被推广以包括参数化、时间依赖性和外力作用。

动力系统 | 机器学习 | 稀疏回归 | 系统识别 | 优化

机器学习(1)和数据科学(2)的进步承诺将带来复杂数据分析和理解的复兴, 从海量多模态数据中提取人类难以掌握的模式。然而, 尽管基于统计关系理解静态数据的工具发展迅速, 从大数据中提炼动态过程的物理模型的进展却很缓慢。这限制了数据科学模型在所采样和构建的吸引子之外推断动态行为的能力。

可以将这一点与开普勒和牛顿的发现进行类比。开普勒凭借当时最全面、最精确的行星数据, 建立了一个以数据为驱动的行星运动模型, 从而得出了著名的椭圆轨道。然而, 这是一种基于吸引子的世界观, 它并未解释产生行星轨道的基本动态关系, 也没有提供一个模型来说明这些天体在受到扰动时会如何反应。相比之下, 牛顿发现了动量和能量之间的动态关系, 这种关系描述了导致这些椭圆轨道的基本过程。这个动态模型可以推广, 用于预测在没有数据采集的情况下的行为。牛顿的模型在工程设计上被证明异常稳健, 使得登陆月球成为可能, 而仅依靠开普勒的模型是不可能实现的。

Bongard 和 Lipson (3) 以及 Schmidt 和 Lipson (4) 的一项开创性突破导致了一种从数据中确定非线性动力系统潜在结构的新方法。该方法使用符号回归 [即遗传编程 (5)] 来寻找非线性微分方程, 并在模型的复杂度 (以项的数量衡量) 与模型精度之间取得平衡。由此得到的模型识别实现了物理学和工程界长期追求的一个目标, 即发现

重要性

理解自然界中的动态约束和平衡促进了知识的快速发展, 并推动了技术进步, 包括飞机、内燃机、卫星和电力。本研究开发了一种新颖的框架, 通过数据测量从简单地发现动力系统的基本控制方程, 利用稀疏技术和机器学习的进展。所得到的模型是简约的, 在模型复杂性与描述能力之间取得平衡, 同时避免过拟合。有许多关键的数据驱动问题, 例如从神经记录中理解认知、推断气候模式、确定金融市场的稳定性、预测和抑制疾病传播, 以及控制湍流以实现更环保的交通和能源。在数据丰富且规律难以捉摸的情况下, 基于数据的动力学发现将在这些努力中继续发挥重要作用。

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The key observation is that for many systems of interest, the function $\mathbf{f}$ consists of only a few terms, making it sparse in the space of possible functions. Recent advances in compressed sensing and sparse regression make this viewpoint of sparsity favorable, because it is now possible to determine which right-hand-side terms are nonzero without performing a combinatorially intractable brute-force search. This guarantees that the sparse solution is found with high probability using convex methods that scale to large problems favorably with Moore's law. The resulting sparse model identification inherently balances model complexity (i.e., sparsity of the right-hand-side dynamics) with accuracy, avoiding overfitting the model to the data. Wang et al. (23) have used compressed sensing to identify nonlinear dynamics and predict catastrophes; here, we advocate using sparse regression to mitigate noise.

To determine the function $\mathbf{f}$ from data, we collect a time history of the state $\mathbf{x}(t)$ and either measure the derivative $\dot{\mathbf{x}}(t)$ or approximate it numerically from $\mathbf{x}(t)$. The data are sampled at several times $t_1, t_2, \dots, t_m$ and arranged into two matrices:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}^T(t_1) \\ \mathbf{x}^T(t_2) \\ \vdots \\ \mathbf{x}^T(t_m) \end{bmatrix} = \begin{array}{c} \text{state} \\ \begin{bmatrix} x_1(t_1) & x_2(t_1) & \cdots & x_n(t_1) \\ x_1(t_2) & x_2(t_2) & \cdots & x_n(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(t_m) & x_2(t_m) & \cdots & x_n(t_m) \end{bmatrix} \end{array} \downarrow \text{time}$$

$$\dot{\mathbf{X}} = \begin{bmatrix} \dot{\mathbf{x}}^T(t_1) \\ \dot{\mathbf{x}}^T(t_2) \\ \vdots \\ \dot{\mathbf{x}}^T(t_m) \end{bmatrix} = \begin{bmatrix} \dot{x}_1(t_1) & \dot{x}_2(t_1) & \cdots & \dot{x}_n(t_1) \\ \dot{x}_1(t_2) & \dot{x}_2(t_2) & \cdots & \dot{x}_n(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ \dot{x}_1(t_m) & \dot{x}_2(t_m) & \cdots & \dot{x}_n(t_m) \end{bmatrix}.$$

Next, we construct a library $\Theta(\mathbf{X})$ consisting of candidate nonlinear functions of the columns of $\mathbf{X}$. For example, $\Theta(\mathbf{X})$ may consist of constant, polynomial, and trigonometric terms:

$$\Theta(\mathbf{X}) = \begin{bmatrix} 1 & \mathbf{X} & \mathbf{X}^{P_2} & \mathbf{X}^{P_3} & \cdots & \sin(\mathbf{X}) & \cos(\mathbf{X}) & \cdots \end{bmatrix}. \quad [2]$$

Here, higher polynomials are denoted as $\mathbf{X}^{P_2}, \mathbf{X}^{P_3}$, etc., where $\mathbf{X}^{P_2}$ denotes the quadratic nonlinearities in the state $\mathbf{x}$:

$$\mathbf{X}^{P_2} = \begin{bmatrix} x_1^2(t_1) & x_1(t_1)x_2(t_1) & \cdots & x_2^2(t_1) & \cdots & x_n^2(t_1) \\ x_1^2(t_2) & x_1(t_2)x_2(t_2) & \cdots & x_2^2(t_2) & \cdots & x_n^2(t_2) \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_1^2(t_m) & x_1(t_m)x_2(t_m) & \cdots & x_2^2(t_m) & \cdots & x_n^2(t_m) \end{bmatrix}.$$

Each column of $\Theta(\mathbf{X})$ represents a candidate function for the right-hand side of Eq. 1. There is tremendous freedom in choosing the entries in this matrix of nonlinearities. Because we believe that only a few of these nonlinearities are active in each row of $\mathbf{f}$, we may set up a sparse regression problem to determine the sparse vectors of coefficients $\Xi = [\xi_1 \ \xi_2 \ \cdots \ \xi_n]$ that determine which nonlinearities are active:

$$\dot{\mathbf{X}} = \Theta(\mathbf{X})\Xi. \quad [3]$$

This is illustrated in Fig. 1. Each column ξ_k of Ξ is a sparse vector of coefficients determining which terms are active in the right-hand side for one of the row equations $\dot{\mathbf{x}}_k = \mathbf{f}_k(\mathbf{x})$ in Eq. 1. Once Ξ has been determined, a model of each row of the governing equations may be constructed as follows:

$$\dot{\mathbf{x}}_k = \mathbf{f}_k(\mathbf{x}) = \Theta(\mathbf{x}^T)\xi_k. \quad [4]$$

Note that $\Theta(\mathbf{x}^T)$ is a vector of symbolic functions of elements of $\mathbf{x}$, as opposed to $\Theta(\mathbf{X})$, which is a data matrix. Thus,

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) = \Xi^T (\Theta(\mathbf{x}^T))^T. \quad [5]$$

Each column of Eq. 3 requires a distinct optimization to find the sparse vector of coefficients ξ_k for the k th row equation. We may also normalize the columns of $\Theta(\mathbf{X})$, especially when entries of $\mathbf{X}$ are small, as discussed in the *SI Appendix*.

For examples in this paper, the matrix $\Theta(\mathbf{X})$ has size $m \times p$, where p is the number of candidate functions, and $m \gg p$ because there are more data samples than functions; this is possible in a restricted basis, such as the polynomial basis in Eq. 2. In practice, it may be helpful to test many different function bases and use the sparsity and accuracy of the resulting model as a diagnostic tool to determine the correct basis to represent the dynamics in. In *SI Appendix, Appendix B*, two examples are explored where the sparse identification algorithm fails because the dynamics are not sparse in the chosen basis.

Realistically, often only $\mathbf{X}$ is available, and $\dot{\mathbf{X}}$ must be approximated numerically, as in all of the continuous-time examples below. Thus, $\mathbf{X}$ and $\dot{\mathbf{X}}$ are contaminated with noise so Eq. 3 does not hold exactly,

$$\dot{\mathbf{X}} = \Theta(\mathbf{X})\Xi + \eta\mathbf{Z}, \quad [6]$$

where $\mathbf{Z}$ is modeled as a matrix of independent identically distributed Gaussian entries with zero mean, and noise magnitude η . Thus, we seek a sparse solution to an overdetermined system with noise. The least absolute shrinkage and selection operator (LASSO) (14, 15) is an ℓ_1 -regularized regression that promotes sparsity and works well with this type of data. However, it may be computationally expensive for very large data sets. An alternative based on sequential thresholded least-squares is presented in Code 1 in the *SI Appendix*.

Depending on the noise, it may be necessary to filter $\mathbf{X}$ and $\dot{\mathbf{X}}$ before solving for Ξ . In many of the examples below, only the data $\mathbf{X}$ are available, and $\dot{\mathbf{X}}$ are obtained by differentiation. To counteract differentiation error, we use the total variation regularization (32) to denoise the derivative (33). This works quite well when only state data $\mathbf{X}$ are available, as illustrated on the Lorenz system (*SI Appendix, Fig. S7*). Alternatively, the data $\mathbf{X}$ and $\dot{\mathbf{X}}$ may be filtered, for example using the optimal hard threshold for singular values described in ref. 34. Insensitivity to noise is a critical feature of an algorithm that identifies dynamics from data (11–13).

Often, the physical system of interest may be naturally represented by a partial differential equation (PDE) in a few spatial variables. If data are collected from a numerical discretization or from experimental measurements on a spatial grid, then the state dimension n may be prohibitively large. For example, in fluid dynamics, even simple 2D and 3D flows may require tens of thousands up to billions of variables to represent the discretized system. The proposed method is ill-suited for a large state dimension n , because of the factorial growth of Θ in n and because each of the n row equations in Eq. 4 requires a separate optimization. Fortunately, many high-dimensional systems of interest evolve on a low-dimensional manifold or attractor that is well-approximated using a low-rank basis Ψ (35, 36). For example, if data $\mathbf{X}$ are collected for a high-dimensional system as in Eq. 2, it is possible to obtain a low-rank approximation using dimensionality reduction techniques, such as the proper orthogonal decomposition (POD) (35, 37).

The proposed sparse identification of nonlinear dynamics (SINDy) method depends on the choice of measurement variables, data quality, and the sparsifying function basis. There is no single method that will solve all problems in nonlinear system identification, but this method highlights the importance of these underlying choices and can help guide the analysis. The challenges of choosing measurement variables and a sparsifying function basis are explored in *SI Appendix, section 4.5 and Appendixes A and B*.

Simply put, we need the right coordinates and function basis to yield sparse dynamics; the feasibility and flexibility of these requirements is discussed in *Discussion* and *SI Appendix section 4.5*

关键的观察是，对于许多感兴趣的系统，函数 f 仅由少数几项组成，使其在可能的函数空间中显得稀疏。近期在压缩感知和稀疏回归方面的进展使这种稀疏性观点变得有利，因为现在可以在不进行组合上不可行的蛮力搜索的情况下确定右侧项哪些是非零的。这保证了使用符合摩尔定律的大规模问题适应性的凸方法能够以高概率找到稀疏解。由此得到的稀疏模型识别天然地平衡了模型复杂性（即右侧动态的稀疏性）与精度，避免了模型对数据的过拟合。Wang 等人 (23) 已经使用压缩感知来识别非线性动力学并预测灾难；在这里，我们提倡使用稀疏回归来减轻噪声。

为了从数据中确定函数 f ，我们收集状态 $x(t)$ 的时间历史，并测量其导数 $\dot{x}(t)$ 或者从 $x(t)$ 数值近似其导数。数据在若干时间 $t_1, t_2, t_3, \dots, t_m$ 处采样，并整理成两个矩阵：

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}^T(t_1) \\ \mathbf{x}^T(t_2) \\ \vdots \\ \mathbf{x}^T(t_m) \end{bmatrix} = \begin{bmatrix} x_1(t_1) & x_2(t_1) & \cdots & x_n(t_1) \\ x_1(t_2) & x_2(t_2) & \cdots & x_n(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(t_m) & x_2(t_m) & \cdots & x_n(t_m) \end{bmatrix} \downarrow \text{time}$$

$$\dot{\mathbf{X}} = \begin{bmatrix} \dot{\mathbf{x}}^T(t_1) \\ \dot{\mathbf{x}}^T(t_2) \\ \vdots \\ \dot{\mathbf{x}}^T(t_m) \end{bmatrix} = \begin{bmatrix} \dot{x}_1(t_1) & \dot{x}_2(t_1) & \cdots & \dot{x}_n(t_1) \\ \dot{x}_1(t_2) & \dot{x}_2(t_2) & \cdots & \dot{x}_n(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ \dot{x}_1(t_m) & \dot{x}_2(t_m) & \cdots & \dot{x}_n(t_m) \end{bmatrix}$$

接下来，我们构建一个由 $\mathbf{X}$ 的列的候选非线性函数组成的库 $\Theta \delta \mathbf{X} \mathbf{D}$ 。例如， $\Theta \delta \mathbf{X} \mathbf{D}$ 可能包含常数、多项式和三角函数项：

$$\Theta(\mathbf{X}) = \begin{bmatrix} 1 & \mathbf{X} & \mathbf{X}^2 & \mathbf{X}^3 & \cdots & \sin(\mathbf{X}) & \cos(\mathbf{X}) & \cdots \end{bmatrix}. \quad [2]$$

这里，更高次的多项式表示为 $\mathbf{X}^2, \mathbf{X}^3$ 等，其中 $\mathbf{X}^2$ 表示状态 x 中的二次非线性项：

$$\mathbf{X}^2 = \begin{bmatrix} x_1^2(t_1) & x_1(t_1)x_2(t_1) & \cdots & x_2^2(t_1) & \cdots & x_n^2(t_1) \\ x_1^2(t_2) & x_1(t_2)x_2(t_2) & \cdots & x_2^2(t_2) & \cdots & x_n^2(t_2) \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_1^2(t_m) & x_1(t_m)x_2(t_m) & \cdots & x_2^2(t_m) & \cdots & x_n^2(t_m) \end{bmatrix}.$$

$\Theta \delta \mathbf{X} \mathbf{D}$ 的每一列代表方程 1 右侧的一个候选函数。在选择这个非线性矩阵中的条目时有很大的自由度。因为我们认为在 f 的每一行中只有少数这些非线性是活跃的，所以我们可以建立一个稀疏回归问题来确定系数的稀疏向量 $\Xi = \frac{1}{2} [\xi_1 \xi_2 \cdots \xi_n]$ ，以确定哪些非线性是活跃的：

$$\dot{\mathbf{X}} = \Theta(\mathbf{X}) \Xi. \quad [3]$$

如图 1 所示。每一列 ξ_k 的 Ξ 是一个稀疏系数向量，用于确定哪些项在方程 1 中某一行方程 $\dot{x}_k = f_k \delta x \mathbf{D}$ 的右侧是活跃的。一旦 Ξ 被确定，就可以如下构建每一行控制方程的模型：

$$\dot{x}_k = f_k(\mathbf{x}) = \Theta(\mathbf{x}^T) \xi_k. \quad [4]$$

请注意， $\Theta \delta x^T \mathbf{D}$ 是 x 元素的符号函数向量，相对于 $\Theta \delta \mathbf{X} \mathbf{D}$ ，后者是数据矩阵。因此，

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) = \Xi^T (\Theta(\mathbf{x}^T))^T. \quad [5]$$

方程 3 的每一列都需要进行不同的优化，以找到第 k 行方程的稀疏系数向量 ξ_k 。我们也可以对 $\Theta \delta \mathbf{X} \mathbf{D}$ 的列进行归一化，尤其是在 $\mathbf{X}$ 的条目较小时，如 SI 附录中所讨论的。

在本文的示例中，矩阵 $\Theta \delta \mathbf{X} \mathbf{D}$ 的大小为 $m \times p$ ，其中 p 是候选函数的数量， $m \gg p$ 是因为数据样本数量多于函数数量；这在受限基（例如公式 2 中的多项式基）中是可能的。在实践中，测试多种不同的函数基并使用所得模型的稀疏性和准确性作为诊断工具来确定表示动态的正确基可能会有帮助。在 SI 附录的附录 B 中，探讨了两个示例，其中稀疏识别算法失败，因为所选基中的动态不稀疏。

实际上，通常只有 $\mathbf{X}$ 可用，并且 $\dot{\mathbf{X}}$ 必须像下面所有连续时间的示例中那样通过数值近似。因此， $\mathbf{X}$ 和 $\dot{\mathbf{X}}$ 都被噪声污染，所以方程 3 并不完全成立。相反，

$$\dot{\mathbf{X}} = \Theta(\mathbf{X}) \Xi + \eta \mathbf{Z}, \quad [6]$$

其中 $\mathbf{Z}$ 被建模为一个由零均值独立同分布高斯条目组成的矩阵，噪声幅度为 η 。因此，我们寻求对带噪的超定系统的稀疏解。最小绝对收缩和选择算子 (LASSO) (14, 15) 是一种 ℓ_1 正则化回归，它促进稀疏性，并且在这类数据上表现良好。然而，对于非常大的数据集，它可能计算成本高。基于顺序阈值最小二乘法的替代方法在 SI 附录的代码 1 中给出。

根据噪声情况，可能需要在求解 Ξ 之前对 $\mathbf{X}$ 和 $\dot{\mathbf{X}}$ 进行滤波。在下面的许多例子中，只有数据 $\mathbf{X}$ 可用，而 $\dot{\mathbf{X}}$ 是通过微分获得的。为了抵消微分误差，我们使用总变差正则化 (32) 对导数 (33) 进行去噪。当仅有状态数据 $\mathbf{X}$ 可用时，这种方法效果相当好，如 Lorenz 系统所示 (SI 附录，图 S7)。或者， $\mathbf{X}$ 和 $\dot{\mathbf{X}}$ 数据也可以进行滤波，例如使用参考文献 34 中描述的奇异值最优硬阈值。对噪声不敏感是从数据识别动力学算法的一个关键特性 (11–13)。

通常，感兴趣的物理系统可以自然地用少数空间变量的偏微分方程 (PDE) 表示。如果数据是从数值离散化或空间网格上的实验测量中收集的，那么状态维度 n 可能会大得难以承受。例如，在流体动力学中，即使是简单的二维和三维流动，也可能需要数万到数十亿个变量来表示离散化系统。所提出的方法不适合大的状态维度 n ，因为 Θ 随 n 的阶乘增长，并且 Eq. 4 中的 n 行方程的每一行都需要单独优化。幸运的是，许多高维感兴趣系统的发展在低维流形或吸引子上，这些可以用低秩基 Ψ (35, 36) 进行良好近似。例如，如果像 Eq. 2 中那样为高维系统收集数据 $\mathbf{X}$ ，则可以使用降维技术获得低秩近似。

所提出的非线性动力学稀疏识别 (SINDy) 方法依赖于测量变量的选择、数据质量以及稀疏化函数基。没有一种方法可以解决非线性系统识别中的所有问题，但该方法强调了这些基础选择的重要性，并可以帮助指导分析。选择测量变量和稀疏化函数基的挑战在《补充信息》附录第 4.5 节以及附录 A 和 B 中进行了探讨。

简而言之，我们需要合适的坐标和函数基以产生稀疏动力学；这些要求的可行性和灵活性在讨论和 SI 附录第 4.5 节中进行了讨论。

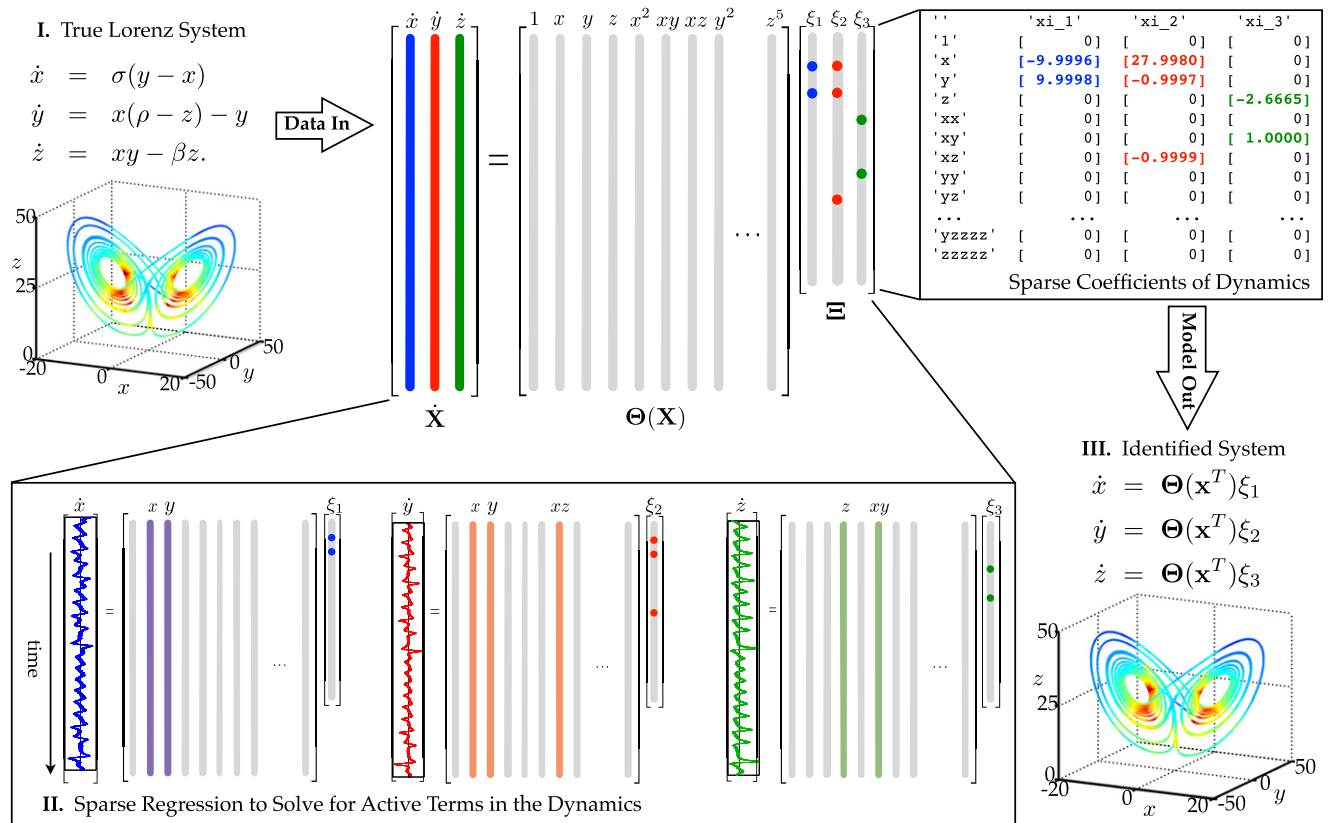


Fig. 1. Schematic of the SINDy algorithm, demonstrated on the Lorenz equations. Data are collected from the system, including a time history of the states $\mathbf{X}$ and derivatives $\dot{\mathbf{X}}$; the assumption of having $\dot{\mathbf{X}}$ is relaxed later. Next, a library of nonlinear functions of the states, $\Theta(\mathbf{X})$, is constructed. This nonlinear feature library is used to find the fewest terms needed to satisfy $\dot{\mathbf{X}} = \Theta(\mathbf{X})\Xi$. The few entries in the vectors of Ξ , solved for by sparse regression, denote the relevant terms in the right-hand side of the dynamics. Parameter values are $\sigma = 10, \beta = 8/3, \rho = 28, (x_0, y_0, z_0)^T = (-8, 7, 27)^T$. The trajectory on the Lorenz attractor is colored by the adaptive time step required, with red indicating a smaller time step.

and Appendixes A and B. However, it may be difficult to know the correct variables a priori. Fortunately, time-delay coordinates may provide useful variables from a time series (9, 12, 38). The ability to reconstruct sparse attractor dynamics using time-delay coordinates is demonstrated in *SI Appendix, section 4.5* using a single variable of the Lorenz system.

The choice of coordinates and the sparsifying basis are intimately related, and the best choice is not always clear. However, basic knowledge of the physics (e.g., Navier–Stokes equations have quadratic nonlinearities, and the Schrödinger equation has $|\mathbf{x}|^2 \mathbf{x}$ terms) may provide a reasonable choice of nonlinear functions and measurement coordinates. In fact, the sparsity and accuracy of the proposed sparse identified model may provide valuable diagnostic information about the correct measurement coordinates and basis in which to represent the dynamics. Choosing the right coordinates to simplify dynamics has always been important, as exemplified by Lagrangian and Hamiltonian mechanics (39). There is still a need for experts to find and exploit symmetry in the system, and the proposed methods should be complemented by advanced algorithms in machine learning to extract useful features.

Results

We demonstrate the algorithm on canonical systems*, ranging from linear and nonlinear oscillators (*SI Appendix, section 4.1*), to noisy measurements of the chaotic Lorenz system, to the unsteady fluid wake behind a cylinder, extending this method to nonlinear PDEs and high-dimensional data. Finally, we show that bifurcation parameters may be included in the models,

recovering the parameterized logistic map and the Hopf normal form from noisy measurements. In each example, we explore the ability to identify the dynamics from state measurements alone, without access to derivatives.

It is important to reiterate that the sparse identification method relies on a fortunate choice of coordinates and function basis that facilitate sparse representation of the dynamics. In *SI Appendix, Appendix B*, we explore the limitations of the method for examples where these assumptions break down: the Lorenz system transformed into nonlinear coordinates and the glycolytic oscillator model (11–13).

Chaotic Lorenz System. As a first example, consider a canonical model for chaotic dynamics, the Lorenz system (40):

$$\dot{x} = \sigma(y - x), \quad [7a]$$

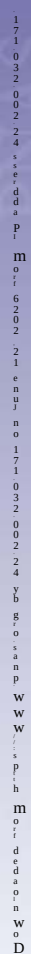
$$\dot{y} = x(\rho - z) - y, \quad [7b]$$

$$\dot{z} = xy - \beta z. \quad [7c]$$

Although these equations give rise to rich and chaotic dynamics that evolve on an attractor, there are only a few terms in the right-hand side of the equations. Fig. 1 shows a schematic of how data are collected for this example, and how sparse dynamics are identified in a space of possible right-hand-side functions using convex ℓ_1 minimization.

For this example, data are collected for the Lorenz system, and stacked into two large data matrices $\mathbf{X}$ and $\dot{\mathbf{X}}$, where each row of $\mathbf{X}$ is a snapshot of the state $\mathbf{x}$ in time, and each row of $\dot{\mathbf{X}}$ is a snapshot

*Code is available at faculty.washington.edu/sbrunton/sparsedynamics.zip.



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of the time derivative of the state $\dot{\mathbf{x}}$ in time. Here, the right-hand-side dynamics are identified in the space of polynomials $\Theta(\mathbf{X})$ in (x, y, z) up to fifth order, although other functions such as sin, cos, exp, or higher-order polynomials may be included:

$$\Theta(\mathbf{X}) = \begin{bmatrix} x(t) & y(t) & z(t) & x(t)^2 & x(t)y(t) & \cdots & z(t)^5 \end{bmatrix}.$$

Each column of $\Theta(\mathbf{X})$ represents a candidate function for the right-hand side of Eq. 1. Because only a few of these terms are active in each row of $\mathbf{f}$, we solve the sparse regression problem in Eq. 3 to determine the sparse vectors of coefficients $\Xi = [\xi_1 \xi_2 \cdots \xi_n]$ that determine which terms are active in the dynamics. This is illustrated schematically in Fig. 1, where sparse vectors ξ_k are found to represent the derivative $\dot{x}_k$ as a linear combination of the fewest terms in $\Theta(\mathbf{X})$.

In the Lorenz example, the ability to capture dynamics on the attractor is more important than the ability to predict an individual trajectory, because chaos will quickly cause any small variations in initial conditions or model coefficients to diverge exponentially. As shown in Fig. 1, the sparse model accurately reproduces the attractor dynamics from chaotic trajectory measurements. The algorithm not only identifies the correct terms in the dynamics, but it accurately determines the coefficients to within .03% of the true values. We also explore the identification of the dynamics when only noisy state measurements are available (SI Appendix, Fig. S7). The correct dynamics are identified, and the attractor is preserved for surprisingly large noise values. In SI Appendix, section 4.5, we reconstruct the attractor dynamics in the Lorenz system using time-delay coordinates from a single measurement $x(t)$.

PDE for Vortex Shedding Behind an Obstacle. The Lorenz system is a low-dimensional model of more realistic high-dimensional PDE models for fluid convection in the atmosphere. Many systems of interest are governed by PDEs (24), such as weather and climate, epidemiology, and the power grid, to name a few. Each of these examples is characterized by big data, consisting of large spatially resolved measurements consisting of millions or billions of states and spanning orders of magnitude of scale in both space and time. However, many high-dimensional, real-world systems evolve on a low-dimensional attractor, making the effective dimension much smaller (35).

Here we generalize the SINDy method to an example in fluid dynamics that typifies many of the challenges outlined above. In the context of data from a PDE, our algorithm shares some connections to the dynamic mode decomposition, which is a linear dynamic regression

(41–43). Data are collected for the fluid flow past a cylinder at Reynolds number 100 using direct numerical simulations of the 2D Navier–Stokes equations (44, 45). The nonlinear dynamic relationship between the dominant coherent structures is identified from these flow-field measurements with no knowledge of the governing equations.

The flow past a cylinder is a particularly interesting example because of its rich history in fluid mechanics and dynamical systems. It has long been theorized that turbulence is the result of a series of Hopf bifurcations that occur as the flow velocity increases (46), giving rise to more rich and intricate structures in the fluid. After 15 years, the first Hopf bifurcation was discovered in a fluid system, in the transition from a steady laminar wake to laminar periodic vortex shedding at Reynolds number 47 (47, 48). This discovery led to a long-standing debate about how a Hopf bifurcation, with cubic nonlinearity, can be exhibited in a Navier–Stokes fluid with quadratic nonlinearities. After 15 more years, this was resolved using a separation of timescales and a mean-field model (49), shown in Eq. 8. It was shown that coupling between oscillatory modes and the base flow gives rise to a slow manifold (Fig. 2, *Left*), which results in algebraic terms that approximate cubic nonlinearities on slow timescales.

This example provides a compelling test case for the proposed algorithm, because the underlying form of the dynamics took nearly three decades for experts in the community to uncover. Because the state dimension is large, consisting of the fluid state at hundreds of thousands of grid points, it is first necessary to reduce the dimension of the system. The POD (35, 37), provides a low-rank basis resulting in a hierarchy of orthonormal modes that, when truncated, capture the most energy of the original system for the given rank truncation. The first two most energetic POD modes capture a significant portion of the energy, and steady-state vortex shedding is a limit cycle in these coordinates. An additional mode, called the shift mode, is included to capture the transient dynamics connecting the unstable steady state (“C” in Fig. 2) with the mean of the limit cycle (49) (“B” in Fig. 2). These modes define the x, y, z coordinates in Fig. 2.

In the coordinate system described above, the mean-field model for the cylinder dynamics is given by (49)

$$\dot{x} = \mu x - \omega y + Axz, \quad [8a]$$

$$\dot{y} = \omega x + \mu y + Ayz, \quad [8b]$$

$$\dot{z} = -\lambda(z - x^2 - y^2). \quad [8c]$$

If λ is large, so that the z dynamics are fast, then the mean flow rapidly corrects to be on the (slow) manifold $z = x^2 + y^2$ given by

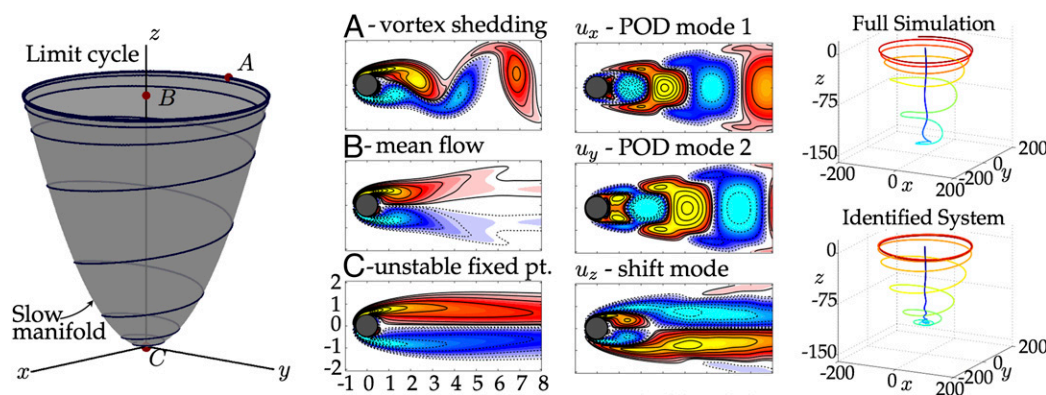


Fig. 2. Example of high-dimensional dynamical system from fluid dynamics. The vortex shedding past a cylinder is a prototypical example that is used for flow control, with relevance to many applications, including drag reduction behind vehicles. The vortex shedding is the result of a Hopf bifurcation. However, because the Navier–Stokes equations have quadratic nonlinearity, it is necessary to use a mean-field model with a separation of timescales, where a fast mean-field deformation is slave to the slow vortex shedding dynamics. The parabolic slow manifold is shown (*Left*), with the unstable fixed point (C), mean flow (B), and vortex shedding (A). A POD basis and shift mode are used to reduce the dimension of the problem (*Middle Right*). The identified dynamics closely match the true trajectory in POD coordinates, and most importantly, they capture the quadratic nonlinearity and timescales associated with the mean-field model.

状态 $\mathbf{x}$ 关于时间的导数。这里，右侧的动力学在多项式空间 $\Theta\delta\mathbf{X}\mathbf{D}$ 中被识别，变量为 $\delta\mathbf{x}, \mathbf{y}, \mathbf{z}\mathbf{D}$ ，最高到五阶，尽管也可以包括其他函数，如 $\sin, \cos, \exp$ 或更高阶多项式：

$$\Theta(\mathbf{X}) = \begin{bmatrix} \mathbf{x}(t) & \mathbf{y}(t) & \mathbf{z}(t) & \mathbf{x}(t)^2 & \mathbf{x}(t)\mathbf{y}(t) & \cdots & \mathbf{z}(t)^5 \end{bmatrix}.$$

$\Theta\delta\mathbf{X}\mathbf{D}$ 的每一列表示方程 1 右侧的一个候选函数。由于在 $\mathbf{f}$ 的每一行中仅有少数这些项是活跃的，我们通过解方程 3 中的稀疏回归问题来确定系数的稀疏向量 $\Xi = \frac{1}{2} [\xi_1 \xi_2 \cdots \xi_n]$ ，从而确定哪些项在动力学中是活跃的。图 1 中以示意方式展示了这一过程，其中稀疏向量 ξ_k 被找到，用以将导数 $\dot{\mathbf{x}}_k$ 表示为 $\Theta\delta\mathbf{X}\mathbf{D}$ 中最少项的线性组合。

在洛伦兹示例中，捕捉吸引子上的动力学能力比预测单条轨迹的能力更为重要，因为混沌会迅速导致初始条件或模型系数的任何微小变化呈指数级发散。如图 1 所示，稀疏模型能够从混沌轨迹测量中准确再现吸引子动力学。该算法不仅识别出动力学中的正确项，而且准确地将系数确定在距真实值 0.03% 的范围内。我们还探讨了在仅有带噪状态测量时的动力学识别（SI 附录，图 S7）。正确的动力学得以识别，并且即使在相当大的噪声值下，吸引子也能保持。在 SI 附录第 4.5 节中，我们使用单个测量 $\mathbf{x}(t)$ 的时延坐标重构了洛伦兹系统中的吸引子动力学。

障碍物后涡脱落的偏微分方程。洛伦兹系统是一个低维模型，用于更逼真的高维偏微分方程模型，以模拟大气中的流体对流。许多感兴趣的系统受偏微分方程 (24) 控制，例如天气与气候、流行病学以及电力网，仅举几例。每个这些例子都具有大数据特征，由大量空间解析测量组成，包括数百万或数十亿状态，并在空间和时间上跨度数个数量级。然而，许多高维的真实世界系统在低维吸引子上演化，使得有效维度大大减小 (35)。

在这里，我们将 SINDy 方法推广到流体动力学中的一个例子，这个例子体现了上文概述的许多挑战。在来自偏微分方程的数据的背景下，我们的算法与动态模态分解有一些联系，后者是一种线性动态回归方法。

(41–43)。数据通过直接对二维纳维-斯托克斯方程 (44, 45) 进行数值模拟收集，用于雷诺数为 100 时流体绕圆柱的流动。通过这些流场测量数据识别出主导相干结构之间的非线性动态关系，而无需了解控制方程。

绕圆柱的流动是一个特别有趣的例子，因为它在流体力学和动力系统中有着丰富的历史。长期以来，人们一直理论认为湍流是随着流速增加而发生的一系列 Hopf 分岔的结果 (46)，从而在流体中产生更丰富和复杂的结构。经过 15 年，流体系统中第一次 Hopf 分岔被发现，在雷诺数为 47 时从稳定层流尾流向层流周期性涡旋脱落的过渡 (47, 48)。这一发现引发了长期的争论，即具有三次非线性的 Hopf 分岔如何在具有二次非线性的 Navier–Stokes 流体中表现出来。再经过 15 年，这个问题通过时间尺度分离和均值场模型得到了解决 (49)，如公式 8 所示。研究表明，振荡模态与基流之间的耦合产生了一个慢流形 (图 2, 左)，由此产生的代数项近似为三次非线性。

这个例子为所提出的算法提供了一个有力的测试案例，因为动力学的潜在形式花了社区中的专家近三十年的时间才被揭示。由于状态维度很大，包括数十万个网格点上的流体状态，因此首先需要降低系统的维度。POD (35, 37) 提供了一个低秩基，从而产生一系列正交模式层次，当截断时，可以在给定秩截断下捕获原系统的大部分能量。前两个能量最高的 POD 模式捕获了能量的显著部分，而稳态涡旋脱落在这些坐标下是一个极限环。为了捕获连接不稳定稳态 (图 2 中的 “C”) 与极限环均值 (49) (图 2 中的 “B”) 的瞬态动力学，增加了一个额外的模式，称为位移模式。这些模式定义了图 2 中的 $\mathbf{x}, \mathbf{y}, \mathbf{z}$ 坐标。

在上述坐标系中，圆柱动力学的平均场模型由 (49) 给出

$$\dot{\mathbf{x}} = \mu\mathbf{x} - \omega\mathbf{y} + A\mathbf{x}\mathbf{z}, \quad [8a]$$

$$\dot{\mathbf{y}} = \omega\mathbf{x} + \mu\mathbf{y} + A\mathbf{y}\mathbf{z}, \quad [8b]$$

$$\dot{\mathbf{z}} = -\lambda(\mathbf{z} - \mathbf{x}^2 - \mathbf{y}^2). \quad [8c]$$

如果 λ 很大，使得 $\mathbf{z}$ 动力学很快，那么平均流会迅速调整到由以下给出的 (慢) 流形 $\mathbf{z} = \mathbf{x}^2 + \mathbf{y}^2$ 上

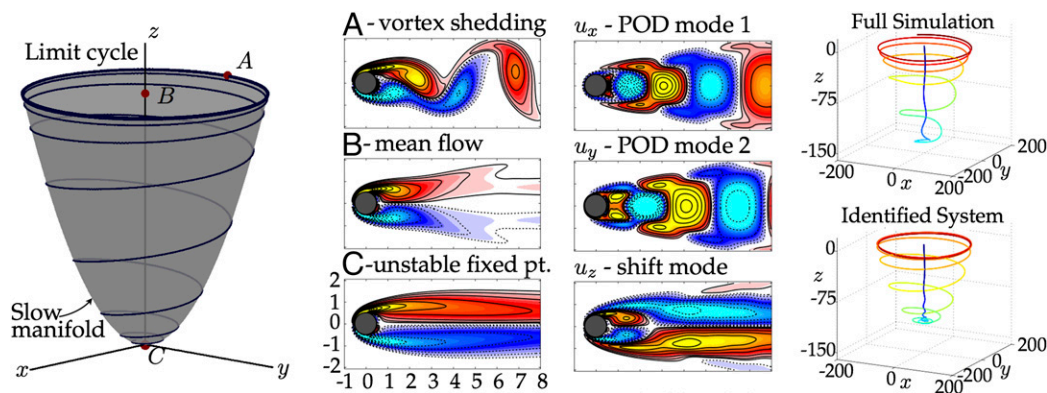


图 2. 来自流体动力学的高维动力系统示例。圆柱体后方的涡脱落是一个用于流动控制的典型例子，与许多应用相关，包括车辆后的阻力减小。涡脱落是 Hopf 分岔的结果。然而，由于 Navier–Stokes 方程具有二次非线性，需要使用具有时间尺度分离的平均场模型，其中快速平均场变形从属于缓慢的涡脱落动力学。图中显示了抛物线型慢流形 (左)，以及不稳定的平衡点 (C)、平均流 (B) 和涡脱落 (A)。使用 POD 基和位移模态来降低问题的维数 (中右)。识别出的动力学在 POD 坐标中与真实轨迹高度匹配，最重要的是，它们捕捉了与平均场模型相关的二次非线性和时间尺度。

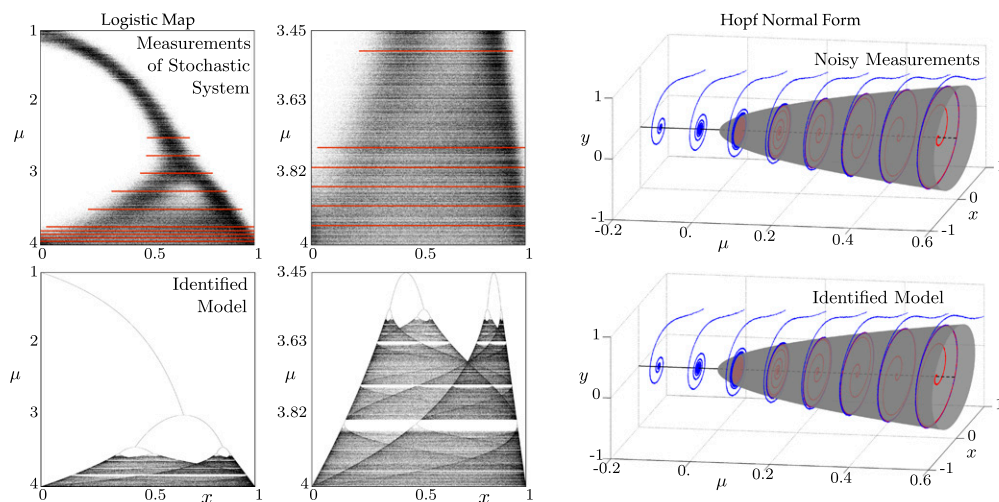


Fig. 3. SINDy algorithm is able to identify normal forms and capture bifurcations, as demonstrated on the logistic map (Left) and the Hopf normal form (Right). Noisy data from both systems are used to train models. For the logistic map, a handful of parameter values μ (red lines), are used for the training data, and the correct normal form and bifurcation sequence is captured (below). Noisy data for the Hopf normal form are collected at a few values of μ , and the total variation derivative (33) is used to compute time derivatives. The accurate Hopf normal form is reproduced (below). The nonlinear terms identified by the algorithm are in [SI Appendix, section 4.4](#) and [Appendix C](#).

the amplitude of vortex shedding. When substituting this algebraic relationship into Eqs. [8a](#) and [8b](#), we recover the Hopf normal form on the slow manifold.

With a time history of these three coordinates, the proposed algorithm correctly identifies quadratic nonlinearities and reproduces a parabolic slow manifold. Note that derivative measurements are not available, but are computed from the state variables. Interestingly, when the training data do not include trajectories that originate off of the slow manifold, the algorithm incorrectly identifies cubic nonlinearities, and fails to identify the slow manifold.

Normal Forms, Bifurcations, and Parameterized Systems. In practice, many real-world systems depend on parameters, and dramatic changes, or bifurcations, may occur when the parameter is varied. The SINDy algorithm is readily extended to encompass these important parameterized systems, allowing for the discovery of normal forms (31, 50) associated with a bifurcation parameter μ . First, we append μ to the dynamics:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}; \mu), \quad [9a]$$

$$\dot{\mu} = 0. \quad [9b]$$

It is then possible to identify $\mathbf{f}(\mathbf{x}; \mu)$ as a sparse combination of functions of $\mathbf{x}$ as well as the bifurcation parameter μ .

Identifying parameterized dynamics is shown in two examples: the 1D logistic map with stochastic forcing,

$$x_{k+1} = \mu x_k (1 - x_k) + \eta_k,$$

and the 2D Hopf normal form (51),

$$\dot{x} = \mu x + \omega y - Ax(x^2 + y^2)$$

$$\dot{y} = -\omega x + \mu y - Ay(x^2 + y^2).$$

The logistic map is a classical model for population dynamics, and the Hopf normal form models spontaneous oscillations in chemical reactions, electrical circuits, and fluid instability.

The noisy measurements and the sparse dynamic reconstructions for both examples are shown in Fig. 3. In the logistic map example, the stochastically forced trajectory is sampled at 10 discrete parameter values, shown in red. From these measurements, the correct parameterized dynamics are identified. The parameterization is accurate enough to capture the cascade of bifurcations as μ is increased, resulting in the detailed bifurcation diagram in Fig. 3. Parameters are identified to within .1% of true values ([SI Appendix, Appendix C](#)).

In the Hopf normal-form example, noisy state measurements from eight parameter values are sampled, with data collected on the blue and red trajectories in Fig. 3 (Top Right). Noise is added to the position measurements to simulate sensor noise, and the total variation regularized derivative (33) is used. In this example, the normal form is correctly identified, resulting in accurate limit cycle amplitudes and growth rates (Bottom Right). The correct identification of a normal form depends critically on the choice of variables and the nonlinear basis functions used for $\Theta(\mathbf{x})$. In practice, these choices may be informed by machine learning and data mining, by partial knowledge of the physics, and by expert human intuition.

Similarly, time dependence and external forcing or feedback control $\mathbf{u}(t)$ may be added to the vector field:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}(t), t),$$

$$i = 1.$$

Generalizing the SINDy algorithm makes it possible to analyze systems that are externally forced or controlled. For example, the climate is both parameterized (50) and has external forcing, including carbon dioxide and solar radiation. The financial market is another important example with forcing and active feedback control.

Discussion

In summary, we have demonstrated a powerful technique to identify nonlinear dynamical systems from data without assumptions on the form of the governing equations. This builds on prior work in symbolic regression but with innovations related to sparse regression, which allow our algorithms to scale to high-dimensional systems. We demonstrate this method on a number of example systems exhibiting chaos, high-dimensional data with low-rank coherence, and parameterized dynamics. As shown in the Lorenz example, the ability to predict a specific trajectory may be less important than the ability to capture the attractor dynamics. The example from fluid dynamics highlights the remarkable ability of this method to extract dynamics in a fluid system that took three decades for experts in the community to explain. There are numerous fields where this method may be applied, where there are ample data and the absence of governing equations, including neuroscience, climate science, epidemiology, and financial markets. Finally, normal forms may be discovered by including parameters in the optimization, as shown in two examples. The identification of sparse governing equations and parameterizations marks a significant step toward the long-held goal of intelligent, unassisted identification of dynamical systems.

We have demonstrated the robustness of the sparse dynamics algorithm to measurement noise and unavailability of derivative

measurements in the Lorenz system (*SI Appendix*, Figs. S6 and S7), logistic map (*SI Appendix*, section 4.4.1), and Hopf normal form (*SI Appendix*, section 4.4.2) examples. In each case, the sparse regression framework appears well-suited to measurement and process noise, especially when derivatives are smoothed using the total-variation regularized derivative.

A significant outstanding issue in the above approach is the correct choice of measurement coordinates and the choice of sparsifying function basis for the dynamics. As shown in *SI Appendix*, Appendix B, the algorithm fails to identify an accurate sparse model when the measurement coordinates and function basis are not amenable to sparse representation. In the successful examples, the coordinates and function spaces were somehow fortunate in that they enabled sparse representation. There is no simple solution to this challenge, and there must be a coordinated effort to incorporate expert knowledge, feature extraction, and other advanced methods. However, in practice, there may be some hope of obtaining the correct coordinate system and function basis without knowing the solution ahead of time, because we often know something about the physics that guide the choice of function space. The failure to identify an accurate sparse model also provides valuable diagnostic information about the coordinates and basis. If we have few measurements, these may be augmented using time-delay coordinates, as demonstrated on the Lorenz system (*SI Appendix*, section 4.5). When there are too many measurements, we may extract coherent structures using

dimensionality reduction. We also demonstrate the use of polynomial bases to approximate Taylor series of nonlinear dynamics (*SI Appendix*, Appendix A). The connection between sparse optimization and dynamical systems will hopefully spur developments to automate and improve these choices.

Data science is not a panacea for all problems in science and engineering, but used in the right way, it provides a principled approach to maximally leverage the data that we have and inform what new data to collect. Big data are happening all across the sciences, where the data are inherently dynamic, and where traditional approaches are prone to overfitting. Data discovery algorithms that produce parsimonious models are both rare and desirable. Data science will only become more critical to efforts in science in engineering, such as understanding the neural basis of cognition, extracting and predicting coherent changes in the climate, stabilizing financial markets, managing the spread of disease, and controlling turbulence, where data are abundant, but physical laws remain elusive.

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